

Performance Under Pressure

Analyzing Formula 1 Driver and Team Performance in Adverse Weather Conditions

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Project Overview

Synopsis

Formula 1 is a global racing series where teams of two drivers compete in the most sophisticated machinery. Over the years, adverse weather driving skill has decided many a championship but has not been studied deeply. This project intends to change that.

Objectives

To examine the relationship between adverse weather conditions and changes in lap time, weather factors with the greatest impact and identify drivers with the most adaptability and performance

Datasets

- FastF1 – Formula 1 Data
- Open-Meteo - Weather Data
- F1 Circuits – Location Data

Approach

Data Preparation

Data was collected from all three sources and subsequently cleaned individually. The race data was combined with the location data and in turn, this data was bolstered with the weather to create a robust dataset for analysis.

Data Analysis

The final dataset was collected and used to understand how drivers and teams perform with different weather conditions. Effects of weather were deeply explored on lap times and race positions.

Constraints

The data was limited to 2018-2025 and therefore, only a picture of modern F1 was visible, not a reflection of the true nature of the sport. In addition, there were several other factors that affect the analysis (pit laps, safety car, etc.) were largely mitigated.

Insights

Driver Performance in Wet Conditions

Using median lap times to benchmark, the results show that Verstappen is approximately 1.2 seconds faster than other drivers in wet weather conditions. Additionally, Verstappen was also the driver with the best average finishing positions in wet weather while Russell was the most improved in wet weather.

Team Performance in Wet Conditions

Teams which are usually lower in the pecking order tend to benefit from the wet weather chaos. Kick Sauber gains around 0.3 seconds on average per lap over other teams in wet weather.

Dry v Wet Driver Performance

There are drivers who finish better in dry or wet weather races, but a majority of drivers are consistent across weather conditions. The data shows that Russell is skilled in the wet while Verstappen is the best across all-weather.

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Project Background & Motivation

Background

Formula 1, often known as F1, is a seasonal global racing series where teams of drivers compete on high-speed circuits with some of the most technologically advanced and optimized cars in motorsport. The winners of these races are often determined by a number of compounding factors; however, weather has been known to be **one of the most unpredictable yet influential elements**.

While Formula 1 teams often invest in optimizing performance under normal racing conditions, adverse weather conditions, such as rain, introduce new challenges that can create **substantial differences in performance**, including reduced grip, limited visibility, and lessened car stability. Formula 1's article, *"Watching the skies - how weather forecasting works in F1"*, highlights the critical role weather plays in race preparation, analyzing the ways teams monitor weather conditions and adjust their strategies accordingly (Formula 1, 2016). Therefore, understanding how weather impacts driver performance can provide valuable insight into a driver's skill and team strategy.

Studies & Reports

A similar study, *"Climate risks in motorsport: Setting boundary conditions in Formula 1"*, analyzes extreme weather patterns such as flooding, heat, and air quality over the course of the 2024 Formula 1 season. This study finds that environmental conditions, particularly extreme heat and heavy rainfall, pose a threat to F1 operations, though it **does not directly address driver and team performance** under these conditions (Orr et al., 2025).

Another study by F1 Metrics, *"Who Was the Best Wet-Weather Driver?"*, used wet-race win rates, crash rates, and points scored to evaluate driver performance in adverse weather conditions. While the study drew conclusions from overall race outcomes, our analysis extends this work by incorporating lap-level insights and weather data to examine how weather impacts performance throughout a race rather than only at its conclusion (F1Metrics, 2014).

Purpose

The purpose of this project is to evaluate the effect of adverse weather conditions on Formula 1 driver and team performance. By integrating historical race weather data from Formula 1 events between 2018 and 2025, this study **analyzes performance at a per-lap level** to measure how changing track conditions influence race outcomes.

Objectives

- 1) Measure the relationship between adverse weather conditions and changes in lap time, sector time, and race position.
- 2) Identify which weather factors have the greatest impact on performance.
- 3) Determine which drivers and teams consistently adapt most effectively to adverse conditions.

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Dataset Sources

Source	Description	Size	Access Methods
FastF1	Our primary dataset will come from the FastF1 Python package, which provides publicly accessible Formula 1 session data from races. This includes lap, weather, and telemetry data for each driver over a 7-year period (2018-2025) returned in Pandas DataFrames. Significant Variables: Lap Data: Driver, Lap Time / Start / Number, Sector Times, Pit Times Weather Data: Rainfall Indicator, Humidity, Track Temp	60.4 MB or ~215,000 total records	https://docs.fastf1.dev/getting_started/installation.html
Open-Meteo	Our secondary data source will be retrieved using the Open-Meteo Historical Weather API. Open-Meteo provides historical weather data using latitude, longitude, date range, and selected weather information variables as parameters. The data returned from the API will be in JSON format, which will be converted and stored as CSV files. Includes hourly data for F1 races from 2018-2025. Significant Variables: Weather Data: Precipitation Amount, Humidity, Temperature	108 KB or 173 total records	https://open-meteo.com/en/docs
F1 Circuits	Our third data set was taken from a publicly accessible GitHub repository by Tomo Bacinger and contains a list of all the F1 circuits and their associated latitude and longitude. It's accessed via a GET request and returns a JSON file. Significant Variables: Circuit Data: Latitude, Longitude, Location, Name	4.94 KB or 40 total records	https://github.com/bacinger/f1-circuits/blob/master/f1-locations.json

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FastF1 | Data Cleaning, Manipulation, & Aggregation

Data Acquisition

Race Data

Race data retrieval was separated into two pipelines: lap and weather; all within a single function, capturing and storing race related data from 2018-2025 using the **FastF1** Python package. Each pipeline configuration contained a set of parameters that would communicate with the **FastF1** API to return the given type of data in the form of a Pandas DataFrame. Each year was stored in a separate datafile for organization.

Issue: Double-header races created duplicate locations within a season, so *EventName* was added to uniquely identify each race.

Data Cleaning

Standardization & Filling Missing Data

The data cleaning process began by joining yearly lap files into a single pandas DataFrame and removing empty columns. Time-based features, including *LapTime*, *Sector1Time*, *Sector2Time*, *Sector3Time*, and *SessionTime*, were converted to timedelta formats to support

comparison and calculation. Missing lap and sector times were then reconstructed from related timing features where possible and estimated when direct calculation was not available.

Location and team name features were standardized across seasons to account for circuit naming, team branding, and ownership changes between 2018 and 2025. Missing *Stint* values were forward-filled by driver and race, while *Compound* values were backfilled to capture missing starting tyre choices. *Speed* features were interpolated to estimate missing values.

Additional missing values were filled using race-specific logic. *Position* gaps appeared only for drivers who did not complete a lap, so missing Position values were assigned as last place. Missing *TrackStatus* values were treated as clear conditions. Finally, *TyreLife* values were completed by extending known values forward and backward within each stint.

Feature Engineering

LapStartTimeUTC

Using the start times for each event from 2018-2025 stored in “*event_location_data.csv*”, a file holding event data from FastF1 sessions, we calculated the UTC time for each lap of the race.

IsPitLap

Using the laps containing pit time values we identified laps involving a pit entry or exit, which can distort lap time performance.

IsTerminalLap

By identify a driver’s max lap number recorded we determined the terminal lap for all drivers, whether due to race completion or retirement.

Missing values were handled grouping by Year, EventName, Driver, and / or LapNumber to preserve the structure of each driver’s session and avoid overlapping races.

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Open-Meteo | Data Cleaning & Manipulation

Data Acquisition

Event & Location Data

Using the [FastF1](#) Python package, we retrieved all Formula 1 race events from 2018–2025, excluding qualifying and other non-race sessions. Because FastF1 does not provide circuit latitude and longitude data, we supplemented the dataset with circuit metadata from the [F1 Circuits GitHub](#) repository. After normalizing location names across both sources, we merged the datasets to create a unified event schedule containing both race timing and location information.

With the location data (latitude, longitude) a query to the [Open-Meteo Archive API](#) (free, no API key required) was done, fetching 14 variables per hour: focusing on precipitation. With a window of 5 hours around each race's start time. This helps to collect weather data missing from the FastF1 data source, such as the amount of precipitation prior to the race and the amount of precipitation during the race.

Feature Engineering

5-Hour Weather Window

With the weather data a –2 h to +2h around each race's UTC start time, the data set of each race was updated adding columns suffixed as `_hm2`, `_hm1`, `_hp0`, `_hp1`, `_hp2`.

Derived Precipitation Features

The precipitation data collected around the start of the race helped to categorize the races, knowing which ones presented residual wet-track risk, how much rain developed during the race and which ones transition from wet to dry and vice versa.

Analytical Derived Features

With the collected data a volatility score was calculated per race. Using the formula $(\text{precip std-dev} \times 5) + \text{temperature std-dev}$. Meaning that a high score represents unpredictable race conditions.

Ground Truth Validation

Tire Compound Cross-Reference

Weather data alone is not enough to truly define if the track was wet at the start of the race, using the lap data collected from [FastF1](#) provides data for the tire compound each driver uses during the first lap, this is better suited for properly knowing if the track remained wet at the beginning of the race if the drivers use a wet or intermediate compound.

Proxy vs Ground Cross Accuracy

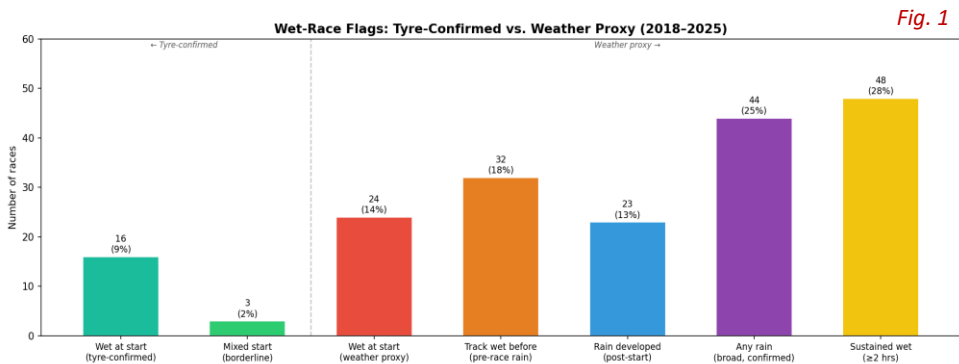
This approach confirmed 16 wet starts, including 3 borderline mixed, the proxy missed 7 races pre-race rain dried at start, but track surface remained damp (Singapore 2022, Turkey 2021...), also the Proxy over-flagged 15 races: drizzle logged but drivers elected slick tires (British GP 2019, Monaco 2019...). ([Figure 1](#))

Output Dataset

This generated 2 data sets, the race weather dataset (173 races x 70+ features). And the wet at start race profiles, which include the confirmed wet at start flags from the cross reference.

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FastF1 & Open-Meteo | Combining Datasets



Hourly-to-Lap Distribution

Source Data

Using the **Open-Meteo** data that delivers one precipitation value per hour (mm/hr), along with the 5 hourly buckets per race, and the **FastF1** lap records which provide the lap start time, rain presence per driver per lap, a calculation of amount of rain per lap was added.

Per-Driver Distribution Formula

The precipitation amount per hour from Open-Meteo was divided by the number of laps with rain fall within that hour, this **interpolation** ensures each driver's exposure tracks their own race pace, not the field aggregate.

Output

New columns with the amount of precipitation per lap as well as the cumulative sum of precipitation to the lap.

Dataset Summary

Coverage

- 172 races covered (2018–2025)
- 189,248 total lap records processed
- 49 races with any precipitation in the 5-hour window
- 37,262 laps with non-zero hourly precipitation rate

Top Wettest Races

- 2024 São Paulo GP
- 2019 German GP
- 2022 Japanese GP
- 2018 Spanish GP
- 2025 British GP

Lap weather profile

Category	Precipitation [mm/hr]	Laps	Percentage
Dry	0	127,510	81%
Light	≤ 0.5	22,611	14%
Moderate	0.5 - 2.0	5,419	3%
Heavy	> 2.0	1,783	1%

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Analyzing Performance in Adverse Conditions | Lap & Sector Times

Rainy Weather Performance

Using the median lap time for each lap as a benchmark, we calculated the average performance of drivers when the track was experiencing rainfall. After filtering out drivers with fewer than 75 wet laps, we averaged the lap time differences. Our results show that Verstappen is, on average, approximately 1.2 seconds faster than other drivers in rainy conditions. (Fig. 2)

Performance Change During Adverse Conditions

We also wanted to examine how driver performance changed when rainy conditions were introduced. Using laps without rainfall as a baseline, we calculated the difference in average lap times during wet conditions.

- **Drivers:** Drivers on the right side of the chart gain the most from wet conditions, those in the middle remain relatively consistent, and those on the left experience a performance decline. (Fig. 4)
- **Teams:** Similarly, we illustrate team performance during adverse conditions by combining the results of both drivers. Teams with the largest gains may benefit from stronger wet weather strategies, while consistent teams may perform well regardless of conditions. (Fig. 3)

Limitations

An important caveat is that rainfall is not the only factor influencing lap times. While weather appears to have a significant impact, other variables may also contribute. To reduce confounding effects, we filtered out pit laps, terminal laps, and laps affected by flags or safety cars.

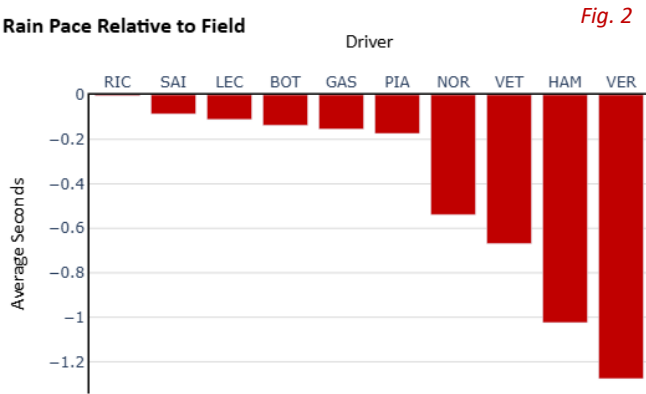


Fig. 2

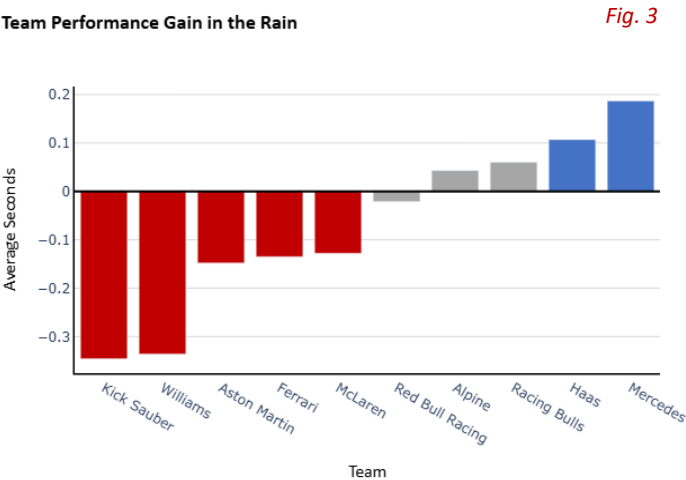


Fig. 3

Driver Performance Gain in the Rain

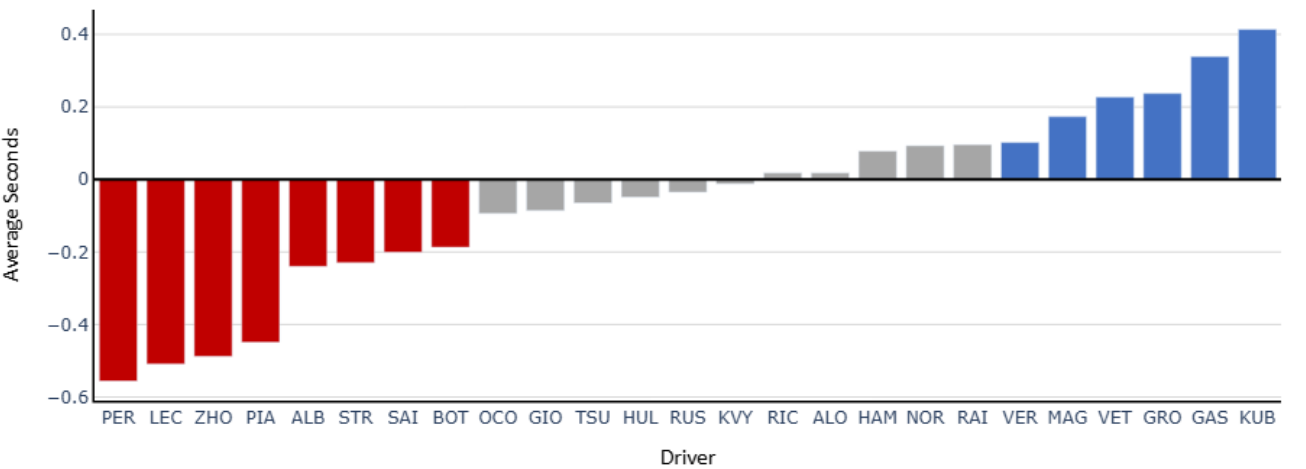


Fig. 4

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Analyzing Performance in Adverse Conditions | Lap Time vs Rain Intensity

Key Findings

Rainy Race Identification (2018–2025)

There are 21 races identified with meaningful rainfall (> 100 clean laps, no SC, VSC, Red flag laps).

Top 3 races with adverse weather conditions and identified rainfall, 2019 German GP (98%), 2022 Japanese GP (87%), 2019 Monaco GP (83%)

Precipitation Intensity Analysis

- WMO bands: Dry / Light ≤ 0.5 / Moderate 0.5–2.0 / Heavy > 2.0 mm/hr
- Heavy rain laps show highest deviation from field median (–0.097 s) indicating widening spread between specialist and struggling drivers
- Dose-response OLS confirms measurable relationship between precipitation rate and lap time deviation from field median. (Figure 5)

Driver Rain Sensitivity Index (RSI)

Using Ordinary Least Squares (OLS) Regression:

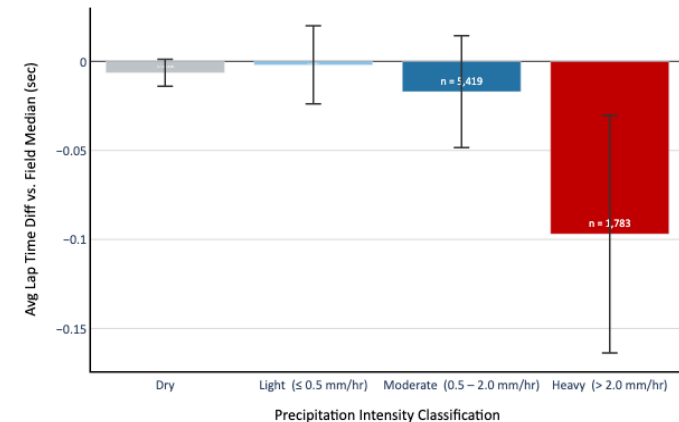
$LapTimeDiff \sim precip_mm$ per driver that drove ≥ 40 wet laps, $p < 0.05$ — 14 drivers qualify

Rain specialists (RSI < 0 , pace improves relative to field) VER, RIC, VET, MAG, PIA.

Rain sensitive RSI > 0 , pace erodes relative to field) PER, LEC, SAI, BOR. (Figure 6)

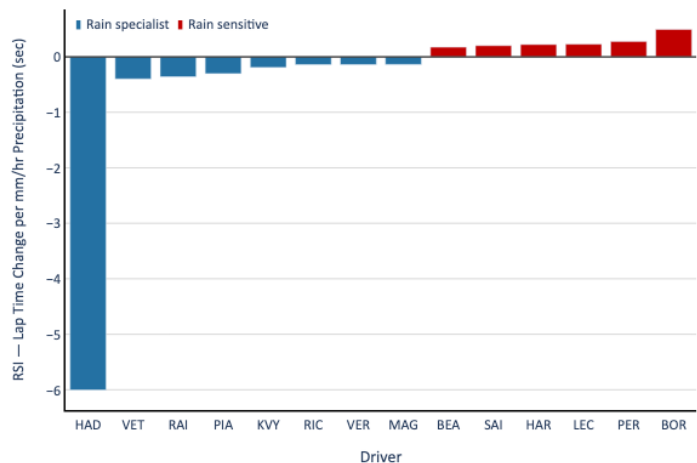
Lap Time Differential by Precipitation Intensity (2018 – 2025)

Fig. 5



Driver Rain Sensitivity Index (2018 – 2025, $p < 0.05$)

Fig. 6



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Analyzing Performance in Adverse Conditions | Driver Performance (Dry v Wet)

Analysis

Methodology

Using the data from **FastF1** and **Open-Meteo**, we create a set of new features to help us compare the performance of drivers (eg: positions by lap, DNFs, wins and podiums, etc.)

Further, all-race data is analysed using a three-pronged approach:

- Race-level
- Driver-level
- Lap-level

The process is then repeated for only wet and dry conditions. This gives us the liberty to draw comparisons amongst drivers in different conditions and create deltas to see their performance change when conditions are different.

We then use this data to create visualizations to identify which drivers can perform better in which conditions.

Visualization

After filtering out drivers with fewer than 5 wet races, we observe the following trends

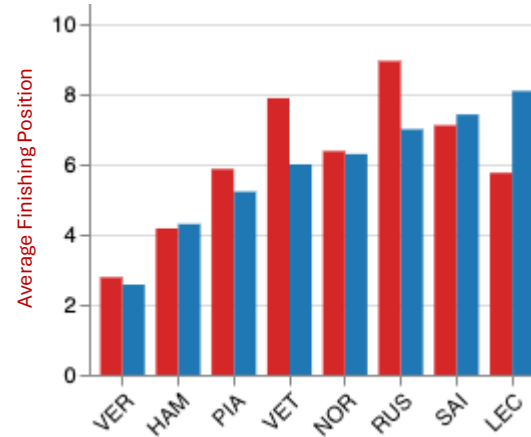


Fig. 7 – Driver v (Dry/Wet) Finishing Position

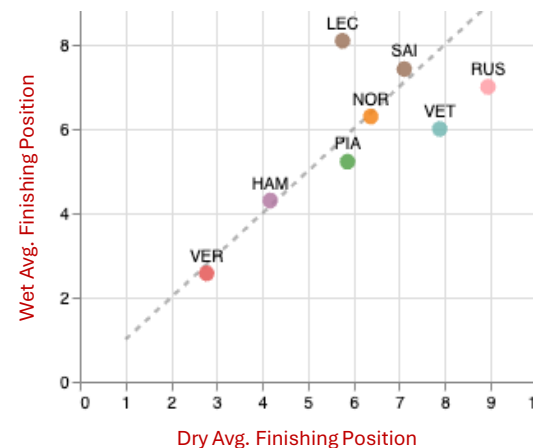


Fig. 8 – Dry v Wet – Avg. Finishing Position

Insights

Driver Wet Weather Finishing Positions

From **Fig. 7**, we can see that Verstappen has the best average finishing position in both wet conditions. This is in line with our earlier observation that Verstappen is faster than other drivers in wet conditions. Coincidentally, Verstappen also has the best average finishing position in dry weather as well.

Driver Dry v Wet Finishing Positions

The diagonal in **Fig. 8** marks the delta of Avg. Finishing Position between dry and wet weather. The implication of this is that if a driver lies on the diagonal, he has similar performance in dry as well as wet weather. Being above the diagonal indicates that a driver is better in the dry than wet and being below the diagonal signals the opposite. The driver with the most negative delta (i. e. Driver who outperforms themselves the most in wet conditions) is Russell. In contrast, the driver who has the worst wet weather performance is Perez (most positive delta).

We can infer that Russell is skilled in wet-weather racing while Verstappen is the best across all-weather conditions.

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References & Statement of Work

References

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Statement of Work

Cody Bauer

- Project Proposal Writeup
- FastF1 Event / Location Data Acquisition
- FastF1 Lap & Weather Data Acquisition
- FastF1 Lap Data Cleaning
- FastF1 Lap Weather Interpolation & Merging
- FastF1 Lap and Sector Time Analysis & Visualizations

Luis Jonathan Godoy Marin

- Open-Meteo / Weather Data Acquisition
- Initial Track Conditions Analysis / Tire Compound Data vs Weather Data Alone
- Precipitation per lap Interpolation & Merging
- Weather Data Manipulation & Cleaning using Lap Data
- Driver Race Sensitivity Analysis & Visualization

Vamsi Krishna Ramesh

- Feature Engineering for Driver Position Analysis
- Race-level, Lap-level and Driver-level Analysis
- Dry vs Wet Race Comparisons and Visualizations
- Repository Instructions and Final Code Review
- Executive Summary and Overall Content Review

Collaboration

Collaboration was facilitated through a series of Zoom calls and Slack chats with team members. Due to the difference in our time zones (EST, PST, and IST), this proved difficult, and there was often a lag in responses. Therefore, we found that meeting on video calls where we could discuss the project in detail often led to better project planning and decision-making.

For future analysis, we believe it would be beneficial to be more proactive in our communication and emphasize video calls over text chat.